

- Specification Version: should say 2.0 (for Windows 11 expectations)

If you don't see TPM 2.0 active, Microsoft's guidance is to enable TPM in firmware (BIOS/UEFI).

Different OEMs label it differently:

- Intel: PTT
- AMD: fTPM

Note: Don't randomly toggle firmware settings if you don't know what they do. But TPM is one of the settings you do want on, because Windows 11 uses it for features like Windows Hello and BitLocker/device encryption.

3) Secure Boot

Think of Secure Boot as Windows' personal security guard or burly bouncer. It helps block nasty boot-level malware. It's also a Windows 11 requirement.

And in 2026 specifically, this matters even more because Microsoft is actively maintaining Secure Boot trust infrastructure (including certificate refreshes rolling via Windows updates ahead of certificate expirations expected later in 2026). Basically, Secure Boot is not a checkbox you tick once and forget. It's an ecosystem. Keep updates on.

Step-by-step: Check whether Secure Boot is ON

Method A: System Information

1. Press Win + R
2. Type msinfo32 → Enter
3. In System Summary, look for:
 - BIOS Mode: should be UEFI
 - Secure Boot State: should be On

If BIOS Mode says "Legacy," that's a sign your system is running old-school boot mode – often linked to older partition styles, older installs, or "somebody upgraded this laptop over time."

Method B: Windows Security

1. Open Start → search Windows Security
2. Go to Device security → check for "Secure Boot" support indicators (varies by build/edition)

Step-by-step: How to get to firmware settings the safe way

Microsoft's recommended path for reaching firmware/UEFI menus from Windows 11 is through Advanced startup.

1. Go to Settings → System → Recovery
2. Under Advanced startup, click Restart now
3. On the blue screen: Troubleshoot → Advanced options
4. Choose UEFI Firmware Settings (if you see it) → Restart

From there, the exact Secure Boot toggle differs by OEM (HP/

Lenovo/Dell/Acer/ASUS all love naming things creatively). If you can't find it quickly, use your laptop brand's support docs.

4) UEFI vs Legacy BIOS

UEFI is the modern firmware interface; Legacy BIOS is the older mode. Windows 11 expects UEFI because it pairs properly with Secure Boot and modern security features.

If you're in Legacy mode, you may still be running Windows 11 depending on how it was installed/upgraded, but you're leaving security – and sometimes compatibility – on the table.

A common "Windows 11 install error" people see is the system not supporting Secure Boot, often tied to being in Legacy mode. Microsoft community guidance commonly points back to switching to UEFI mode (and ensuring TPM is enabled).

Note: Switching Legacy → UEFI sometimes also involves disk partition style changes (MBR → GPT). That's an advanced operation. We'll cover clean reinstall and recovery later in the book. For Chapter 1, your job is simply to identify where you stand, not do risky surgery on Day One.

5) Confirm Windows is genuine

In India, let's not pretend this isn't a thing. A surprising number of PCs run "activated somehow" Windows images that behave strangely: updates fail, Microsoft Store breaks, security features misbehave, and recovery becomes unreliable.

Microsoft's activation help pages start with the basics: verify you're running Windows 11 and check your version/build (yes – winver again).

Step-by-step: Check activation status

1. Go to Settings → System
2. Scroll to Activation
3. You want to see: Active / "Windows is activated"

If activation is failing, Windows includes an Activation troubleshooter that Microsoft explicitly recommends for common errors.

Note: If your Windows isn't genuine, fix that first. Every other optimisation or troubleshooting step becomes a gamble.

6) OEM bloatware

Most consumer laptops ship with a small parade of pre-installed apps:

- trial antivirus
- vendor "care" suites
- cloud storage prompts
- shopping apps
- game launchers you never asked for
- "optimisers" that duplicate what Windows already does

The goal isn't to delete everything. The goal is to remove anything you don't use that:

- runs at startup,